# ASSESSMENT 1

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| ***Deadline:*** | Hand in by midnight 22 August. |
| ***Evaluation:*** | **20% of your final course grade.** |
| ***Late Submission:*** | Refer to the course guide. |
| ***Work*** | This assignment may be done in **pairs** or **individually**. |
| ***Purpose:*** | Learning outcomes 1 and 2 from the course outline. |

## PROJECT DESCRIPTION

In this class project, our objective is to utilise a Kaggle competition as a practical learning experience. We will focus on exploring different machine learning algorithms, including AdaBoost, as well as various techniques to get the most out of our results. Throughout the project, we will also gain familiarity with AWS technologies, allowing us to gain familiarity with cloud technologies.

The target problem and dataset for this assignment and class competition is to predict whether an applicant is approved for a loan https://www.kaggle.com/competitions/playground-series-s4e10/submissions

Unfortunately, there are no suitable running competitions that meet the requirements, so this Kaggle competition has formally closed, but you can still submit your answers and receive submissions scores – you will however not be added to the leaderboard. We just need the submission scores for our purposes, so this is not a problem.

The project has 5 tasks. Your first step should be to create a Kaggle account and acquire the relevant dataset.

Make sure you submit a tidy, structured notebook that follows the template, having the preamble completed, alongside all relevant headers and interpretations/analyses. Include in the introduction your Kaggle screenshot of your best AdaBoost submission score, and what your class competition ranking was at the time of submission so we can verify it. Also, your best AdaBoost submission.csv should be included in your zipped file so we can verify your result too.

### HAND-IN:

Make sure that all the outputs are embedded in your notebook and submit both the notebook and the html and PDF versions of your assignment (yes - 3 file formats). Submit also your best submission.csv file which generated your best AdaBoost submission score for the competition**. NO AI-generated code allowed – you must also submit your Generative AI use statement.**

**Use a project template provided on Stream and describe at the top all the steps you have performed.**

Submit just 1 notebook please if possible, but maximum of 2 if you need to include AWS components separately.

# TASK 1: ADABOOST IMPLEMENTATION (25 MARKS)

You can complete the following tasks by either using the synthetics dataset found in the notebooks, or you might want to select 2 features from the Kaggle dataset. I would recommend the latter. You must make sure you work with the AdaBoost code from the class. Completely different implementations of AdaBoost having a significant departure from the class code will not count in this component.

1. Implement AdaBoost Algorithm: (5 marks)
    a. Properly implement the AdaBoost algorithm from class code, then test it and verify it works.
2. Explore Margins and Analyse Results: (5 marks)
    a. Examine how margins are related to classifier performance and the overall confidence/performance of the model.
3. Adjust Weak Classifier Type: (5 marks)
    a. Conduct experiments with different weak classifier types within the AdaBoost ensemble.
    b. Observe how the choice of weak classifier impacts the overall performance of the model.
4. Focus on Difficult Examples / Adjust Classification Threshold: (5 marks)
    a. Investigate techniques to make AdaBoost focus more on challenging-to-classify examples.
    b. Analyse the effect of emphasizing these examples on the accuracy and generalization of the model.
    c. Explore the impact of adjusting the classification threshold for decision-making.
    d. Examine how threshold adjustments affect precision, recall, and the overall behaviour of the model.
5. Implement an ensemble pruning/optimisation procedure that is executed after the model training (5 marks)
    a. Try out one or more techniques that strive to optimise the AdaBoost ensemble and test to see if they work. Your task here is essentially to see if you can remove some weak classifiers from the ensemble in a principled way so as to improve the overall ensemble performance.

# TASK 2: ADABOOST IN-CLASS KAGGLE COMPETITION (35 MARKS)

In this class, we will be conducting an in-class competition to determine the student who achieves the best solution using **our class implementation of AdaBoost**. You must use your implementation of AdaBoost based on the class notebooks, which potentially features some of your modifications/enhancements as well as the required adaptation to work with multiple features. AdaBoost implementations from scikit-learn or other modules are not valid for this. Again, you must make sure you work with the AdaBoost code from the class. Completely different implementations of AdaBoost from the class code will not count in this component.

1. Competition Score Calculation: (20 marks)
    a. The competition score for each student's (or pair of students) submission will be based on their model's performance on a held-out test dataset score as generated by Kaggle.
    b. Each student must enter their best score calculated by the Kaggle platform on a Google Sheet document: https://docs.google.com/spreadsheets/d/1nDXLVp6OnJh3LAKsUhS5vID0NcysVdeS7ZkITbMj_48/edit?gid=0#gid=0
    c. Evidence in the form of a screenshot from Kaggle must be embedded in the notebook for the best score achieved, and provide evidence that your implementation of AdaBoost generated the submission.csv file of your best result.
    d. The winning student(s) will receive 20 marks for obtaining the top-performing solution from our class.
    e. The student with the next best solution will be awarded 17 marks. Subsequent rankings will receive one less mark than the preceding position, with the minimum being 10 marks for a successful submission.
    f. Your first Kaggle submission must be made by July 31 and your initial score must be entered into the class competition Google Sheet by this date, otherwise 5 marks will be deducted.
2. Tuning AdaBoost and the Training Procedure (15 marks + potentially 5 bonus marks for exceptional work)
    a. Perform robust model calibration and demonstrate its effects
    b. Perform tuning on the AdaBoost algorithm to optimise its performance where possible.
    c. Utilise cross-validation techniques to ensure robustness and avoid overfitting.

d. Conduct feature engineering/selection to identify the most relevant and informative features for the task.
e. Exploration of the effects of other modifications/enhancements to AdaBoost.
f. Analyse and report how the performance of the model changes with different feature subsets.

## TASK 3: KAGGLE COMPETITION USING A RANGE OF ALGORITHMS (10 MARKS)

1. Explore a Range of Different Algorithms
2. Demonstrate Hyperparameter Tuning where necessary/possible.
3. Perform feature engineering and feature selection
4. Document accuracy over time
    a. Create visualizations, such as line plots, to display the accuracy trend of various experiments.

Make sure that you screenshot and embed into your notebook your best score and your position on the Kaggle leaderboard.

## TASK 4: AWS TECHNOLOGY USAGE AND COURSE COMPLETION(15 MARKS)

1. AWS Demonstration: (10 marks)

In this task, you will showcase your proficiency in utilising AWS technologies covered thus far in the course. The focus will be on applying AWS services to enhance machine learning workflows, leveraging cloud computing capabilities for scalable and efficient data processing (even when the datasets are small like this one). The key AWS technologies to consider demonstrating are as follows:

1. Amazon SageMaker:
2. Amazon S3 (Simple Storage Service):
3. AWS Glue:
4. Hyperparameter Tuning:
5. Amazon SageMaker Autopilot:

However, not all the above need to be used.

2. AWS Course Completion: (5 marks)

Include a screenshot of your AWS Academy course progression and embedding the image into the notebook, showing how far you have progressed in your AWS Academy courses. Without this, zero marks will be awarded for Task 4.

## TASK 5: FILLING OUT THE JUPYTER NOTEBOOK TEMPLATE (15 MARKS)

In the top cell of your notebook, you must complete a checklist/preamble in markdown that confirms you have addressed each of the major components of the assignment. For every requirement listed, you must provide a working internal hyperlink that directs the marker to the exact section of your notebook where you have completed the corresponding work. This will allow us to verify your work efficiently and is essential to the marking process. You are expected to be explicit and accurate with your links. Incomplete or vague references will result in penalties.

This part of the assignment is not just administrative — it is designed to prompt you to reflect critically on the completeness and quality of your work. This checklist represents your own evaluation of how well you have met the task objectives, and is an important part of the learning outcomes. Being able to map your work to specific deliverables is a key skill in both academic and professional settings, and the process of doing so supports your ability to evaluate your own progress and performance. You should approach this section with care and attention, and treat it as a formal declaration of what you have done, what you are submitting, and how it aligns with the assessment criteria. Use the notebook markdown and internal linking syntax properly, and ensure that the linked content is clearly presented and corresponds precisely to the task described.